Respiratory Assessment

NURA 330 Health Assessment | Module 3 Lecture Study Guide

Exam 1 content · reviewed for the final

Lecture roadmap: Case: James → A&P review (airways, lobes, alveoli, diaphragm, pleura, respiratory drive) → Adult vs. pediatric airway → Subjective assessment and SDOH → Pack-years → Objective assessment: vital signs, inspection, cyanosis across skin tones, retractions, auscultation technique, normal and adventitious breath sounds, palpation → Paradoxical breathing and barrel chest → Labs and chest X-ray → What to report → Lifespan → Case wrap-up → Documentation

Exam 1 blueprint: Module 3 was worth **20–22 questions**, the largest section. Know: lung lobes, pediatric vs. older adult differences, pack-years, retraction types, 1:2 vs. 1:1 AP ratio, paradoxical breathing, ladder-pattern auscultation to T10, normal breath sound locations, every adventitious sound, key terms, red flags, and objective charting.

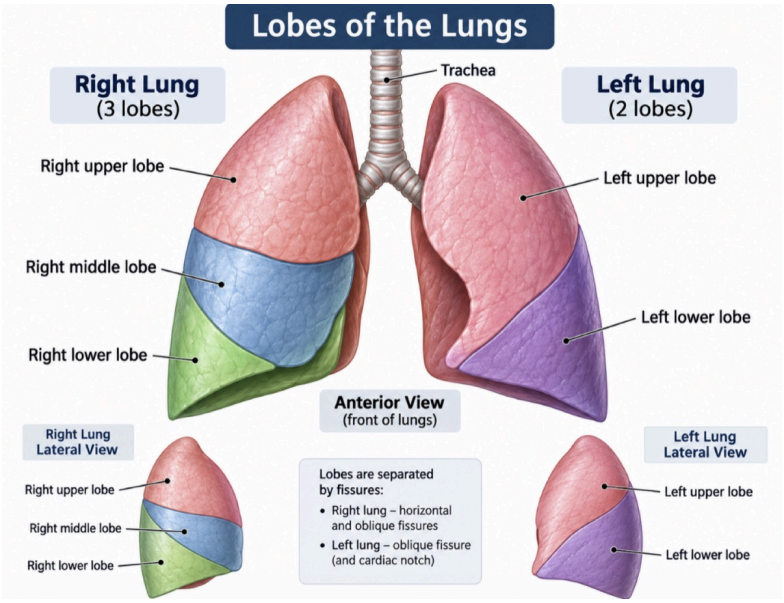
How to read this guide: Everything without a tag comes from the lecture slides or the slide notes. Anything else is tagged: STUDY GUIDE emphasized on the exam study guide but not spelled out on the slides; ADDED information not in the slides; CLARIFIED a slide point explained further; CORRECTED an error or inconsistency in the slides. Every tag is also listed in the notes at the end.

Case: James, 68. Admitted with increasing shortness of breath and cough; history of hypertension and a **30 pack-year** smoking history. Sitting upright: "I just feel like I can't catch my breath. It seems worse today." Pauses while speaking; breathing fast. **HR 104 | RR 26 | BP 148/82 | SpO_2 90% RA | T 38.1 °C (100.6 °F)**. Before touching him, you already notice: high RR, low SpO_2, upright positioning, trouble speaking in full sentences, fever. We return to James at the end.

Anatomy and Physiology Review

Structure	Key points
Upper airway	Nose, mouth, pharynx, larynx: **filters, warms, and humidifies** air
Lower airway	Trachea → bronchi → bronchioles → alveoli
Lungs	**Right lung: 3 lobes** (upper, middle, lower). **Left lung: 2 lobes** (upper, lower), leaving room for the heart

Structure	Key points
Alveoli	Tiny air sacs where **O_2 and CO_2 gas exchange** happens
Diaphragm	**Primary muscle of breathing**; contracts and moves **down** on inspiration
Accessory muscles	Recruited when breathing takes more effort; **using them is an important assessment finding**
Pleura	Membranes around the lungs that let them move smoothly



Lobes of the lungs (lecture slide)

- **Normal breathing needs:** a patent airway + adequate ventilation + adequate perfusion + effective gas exchange.
- **Respiratory drive** is regulated mainly by the **brainstem**, responding to changes in CO_2, O_2, and pH.

ADDED **Accessory muscles to know:** sternocleidomastoid, scalenes, and trapezius (inspiration), plus the abdominal muscles (forced expiration).

Lifespan Differences

Infants & children	Older adults
Larger head; **larger tongue** Infants are **obligate nose breathers**; smaller nares More flexible trachea; **narrow airway** (small diameter) Faster respiratory rates than adults Abdominal breathing is common Watch for retractions, nasal flaring, grunting **Distress can progress quickly**	Decreased lung elasticity (compliance) and chest expansion **Weaker cough** and respiratory muscles **Higher risk of infection and aspiration** Illness may present with **subtle** symptoms

CLARIFIED **Why it matters:** A small amount of swelling or mucus narrows a child's small airway much more than an adult's, and a stuffy nose matters for an infant who breathes mainly through the nose. In older adults, confusion or new fatigue may be the first sign of pneumonia, before fever or cough.

Subjective Assessment

Start with "What is troubling you most at the moment?" then ask about:

- Diagnosed respiratory conditions: asthma, COPD, pneumonia, allergies
- **Home oxygen**; home equipment (CPAP, BiPAP, nebulizer)
- Medications, herbs, supplements for breathing
- Breathlessness; **cough** (and its characteristics); changes in breathing; new shortness of breath with activity
- Energy level and changes
- Smoking or vaping
- STUDY GUIDE **Pleuritic chest pain**: sharp pain that worsens with a deep breath or cough

Expand on pertinent positives with OLDCARTS.

Psychosocial and SDOH

Tobacco / vaping	Current or past use, secondhand smoke, readiness to quit
Home and environment	Mold, poor air quality, wood smoke, allergens, pets, **occupational dust or chemicals**
Housing	Stability; access to heat, clean air, electricity (needed for oxygen concentrators and nebulizers)
Access to care	Primary care, transportation, insurance; ability to get medications, inhalers, oxygen, equipment
Health literacy	Understanding of the condition; **correct inhaler and oxygen use**
Psychosocial	**Anxiety can worsen the feeling of dyspnea**; effects on sleep, work, quality of life; support system

Pack-Year History

Packs per day × Years smoked = Pack-years

Based on 20 cigarettes per pack. Risk of smoking-related disease is **dose-dependent**: it rises in proportion to cumulative pack-years.

ADDED **Practice:** 2 packs/day × 15 years = **30 pack-years** (like James). ½ pack/day × 20 years = **10 pack-years**. 10 cigarettes/day = 0.5 packs.

Objective Assessment

Order: **vital signs** → **inspect** → **auscultate** → **palpate** → **document**. Always start with the least invasive.

Vital Signs

Respiratory rate	Rate, rhythm, depth, effort. **Often an early indicator of deterioration**
SpO_2	Oxygenation; consider baseline and presentation
Heart rate	Tachycardia may occur with hypoxemia, distress, fever, anxiety
Temperature	Fever may indicate a respiratory infection
Blood pressure	Can change with significant distress or deterioration

Inspection

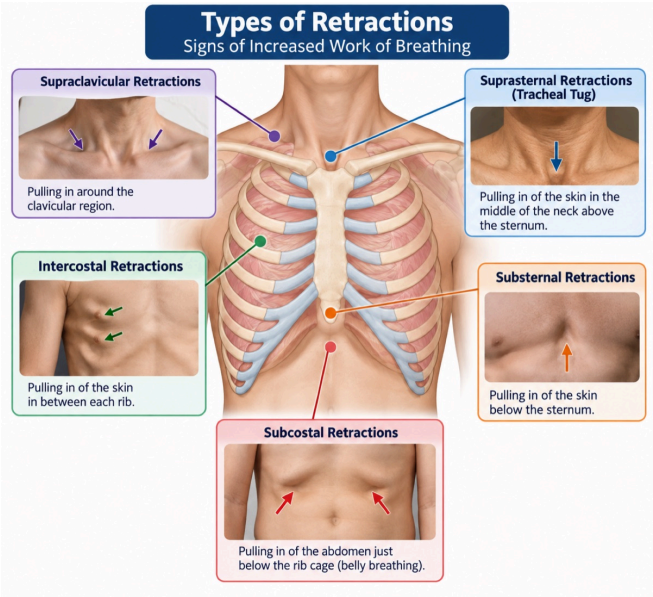
What do you see when you walk in? Respiratory **effort** and **rate** (tachypnea); accessory muscle use and **retractions**; **nasal flaring**; **head bobbing**; color of skin, lips, hands, and feet; chest shape and **symmetric expansion**; level of consciousness; speech (full sentences?).

Cyanosis Across Skin Tones

- **Lighter skin:** blue or purple tint around lips and nail beds.
- **Darker skin:** may look **gray, ashen, or dull**.
- **Most reliable sites** (least pigment): **tongue, oral mucosa, inner lips, hard palate** (central cyanosis); also conjunctivae, nail beds, palms, soles.
- **Always confirm with pulse oximetry.** Visual inspection alone can be misleading.

Signs of Increased Work of Breathing: Retractions

Type	Where the skin pulls in
Subcostal	Abdomen just **below the rib cage** ("belly breathing")
Intercostal	**Between the ribs**
Suprasternal (tracheal tug)	Middle of the neck **above the sternum**
Supraclavicular	**Above the clavicles**
Substernal	**Below the sternum** (at the xiphoid) CORRECTED see note 1



Types of retractions (lecture slide)

Chest Shape and Movement

	Normal chest	Barrel chest
AP : lateral ratio STUDY GUIDE	**1:2** (front-to-back depth is about half the side-to-side width)	**1:1** (as deep as it is wide; rounded)
Cause	n/a	Chronic lung **hyperinflation** from air trapping; common in **COPD, especially emphysema**
Often seen with	n/a	Dyspnea, increased work of breathing

Paradoxical breathing: the chest and abdomen move **opposite** to normal (one moves in while the other moves out), instead of both rising on inspiration. May mean **respiratory muscle fatigue**, **chest trauma (flail chest)**, or diaphragm weakness or paralysis. It signals significant distress: check work of breathing, SpO_2, and mental status, and **escalate promptly** if new.

Palpation

Evaluates **chest expansion, depth of breathing, and symmetry**: how well the lungs expand and recoil.

ADDED **How to do it:** Place your hands on the posterior chest at about the 10th rib with thumbs near the spine, pinching up a small fold of skin. As the patient inhales deeply, your thumbs should move apart **equally**. Unequal movement (asymmetrical expansion) is a red flag.

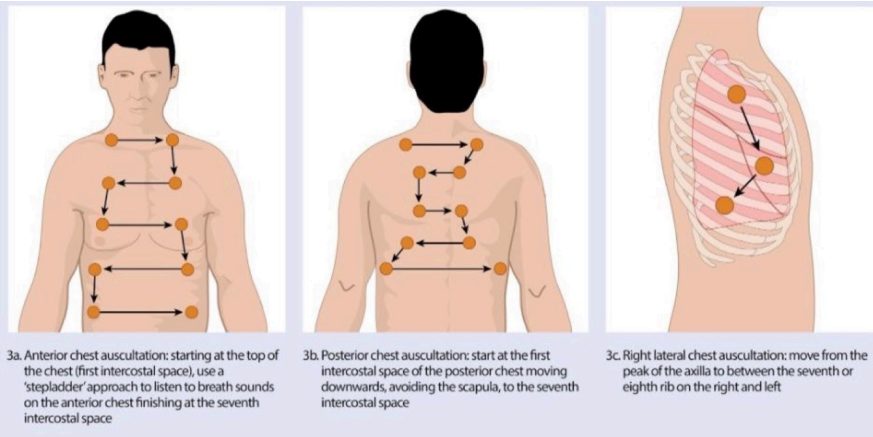
Auscultation

Technique

- Use the **diaphragm**, **directly on bare skin**.
- Patient breathes **slowly and deeply through an open mouth**.
- Listen to **one full inspiration and expiration** at each spot.
- Go **top to bottom**, **comparing side to side** at each level (the "**ladder**" pattern).
- Cover **anterior, lateral, and posterior** fields; avoid listening over bone or the scapulae.
- Note the **location, timing, and quality** of any abnormal sound, and interpret it with the whole assessment.

Field	Where
Anterior	Apices **above the clavicles** down to about the **6th rib**
Lateral	Along the **midaxillary lines**, including lower fields. **The most commonly missed area**
Posterior	Apices **above the scapulae** down to about **T10**. **Most lung tissue is posterior**; do not skip the back

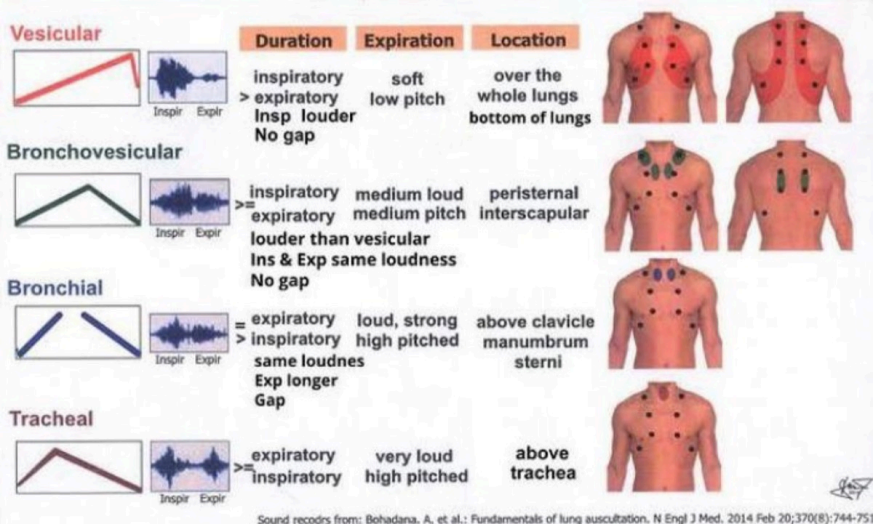
Teaching pearl: the **right middle lobe** (and left lingula) is best heard **anteriorly/laterally**; the **lower lobes** are mainly heard **posteriorly**.



Anterior, posterior, and lateral auscultation sequence (lecture slide)

Normal (Expected) Breath Sounds

Sound	Where	Quality
Bronchial	Over the **trachea and larynx**	**High-pitched, loud**; expiration longer than inspiration
Bronchovesicular	Over the **major bronchi** (beside the sternum in front; between the scapulae in back)	**Medium** pitch; inspiration = expiration
Vesicular	Over **most of the lung surface** (peripheral fields)	**Low-pitched, soft, rustling**; inspiration longer than expiration



Normal breath sounds by location (lecture slide)

CLARIFIED A normal sound heard in the **wrong place** is abnormal. For example, bronchial sounds over the peripheral lung fields can mean consolidation (e.g., pneumonia).

Adventitious (Abnormal) Breath Sounds

Sound	What it sounds like	When	What it suggests
Wheezing	High-pitched, musical	Usually **expiration**	**Narrowed airways** (asthma, COPD)
Crackles (rales)	Clicking, rattling, popping. **Fine:** soft, high, brief. **Coarse:** louder, lower, longer	Usually **inspiration**	**Fluid in the alveoli** (pneumonia, heart failure, pulmonary fibrosis)
Stridor	Harsh, high-pitched	**Inspiration**	**Upper airway obstruction** (croup, foreign body). **An emergency**
Rhonchi	Low-pitched, snoring	Usually expiration	**Secretions in large airways** (bronchitis); may clear with coughing
Pleural friction rub	Grating	**Both** inspiration and expiration	**Inflamed pleura** rubbing together (pleuritis)

Sound	What it sounds like	When	What it suggests
Grunting	Low-pitched, short	Expiration	Respiratory distress, especially in infants

Always ask about the patient's **baseline** (e.g., someone with COPD may always have some wheezing).

Further Investigation

Test	Why
CBC	WBC for infection; Hgb/Hct for oxygen-carrying capacity
ABG	Oxygenation, ventilation, acid-base balance
BMP/CMP	Electrolytes and kidney function
Sputum culture	Identifies the organism causing infection
Blood cultures	Suspected serious or systemic infection
BNP	Helps identify heart failure as a cause of dyspnea or pulmonary edema
Chest X-ray	Pneumonia, pulmonary edema, pleural effusion, pneumothorax, hyperinflation; always correlate with the assessment

What to Report Immediately

Red flags:

- **SpO_2 below 92%** (or below the patient's baseline)
- Pain (e.g., pleuritic chest pain)
- **Worsening dyspnea**
- **Decreased LOC, restlessness, anxiety, irritability** (early signs of hypoxia)
- STUDY GUIDE **Acute stridor** and **asymmetrical chest expansion**

Medical Terminology

Term	Meaning
Tachypnea / Bradypnea	Abnormally fast / slow breathing
Dyspnea	Difficult or uncomfortable breathing
Orthopnea	Difficulty breathing when lying flat
Productive / Non-productive cough	Produces sputum / dry, no sputum
Wheezing, rales (crackles), rhonchi, stridor	See the adventitious sounds table
Adventitious breath sounds	Abnormal or extra breath sounds
SpO_2	Percentage of hemoglobin carrying oxygen
Pulmonary edema	Fluid accumulation in the lungs

Back to James

Focused assessment findings: sitting upright, mildly anxious; RR 26, shallow; **accessory muscle use**; chest expansion symmetric; **productive cough with thick yellow sputum × 3 days**; **right-sided chest pain with deep breaths or coughing**; **crackles in the right lower lobe**, other fields clear; SpO_2 90% RA; "I feel like I have to work harder to breathe."

What do you see?	What does it mean?	What will you do?
RR 26, SpO_2 90%, accessory muscles, crackles RLL, productive yellow sputum, fever, pleuritic pain, short phrases, HR 104	Impaired oxygenation and increased work of breathing, consistent with a possible infectious/consolidation process in the right lower lobe (no medical diagnosis needed)	Position upright; oxygen per orders/protocol; monitor SpO_2 and respiratory status; reassess; **communicate concerning findings to the provider**

Documentation

Include: **rate, rhythm, depth**; **breath sounds** and location; accessory muscle use or labored breathing; **cough** (productive or not; sputum color and consistency); **SpO_2** and oxygen source. Use precise numbers and terms; never "breathing fine." Document changes promptly.

Vague (avoid)	Objective (chart this)
"Pt breathing fast and coughing a lot."	"Respirations 26/min, labored. Productive cough with thick yellow sputum. Breath sounds diminished bilaterally with expiratory wheezes. O_2 sat 91% on room air."
"Pt breathing a little fast and says she's wheezing."	"Respirations 20/min, even and unlabored. Breath sounds clear in upper lobes, faint expiratory wheezes at bases. O_2 sat 95% on room air. Cough occasional, nonproductive."

Quick Review

Topic	Key points
Lobes	Right 3, left 2
Pediatric airway	Narrow, bigger tongue, obligate nose breathers, fast deterioration
Older adult	Less compliance, weaker cough, aspiration and infection risk, subtle signs
Pack-years	Packs/day × years
Retractions	Sub**costal**, inter**costal**, supra**sternal** (tracheal tug), supra**clavicular**, sub**sternal**
AP ratio	Normal 1:2; barrel chest 1:1 (COPD/emphysema)
Auscultation	Diaphragm, bare skin, open-mouth deep breaths, ladder pattern side to side; anterior to 6th rib, lateral, posterior to T10
Normal sounds	Bronchial = trachea; bronchovesicular = major bronchi; vesicular = periphery
Adventitious	Wheeze = narrow airway; crackles = alveolar fluid; stridor = upper airway emergency; rhonchi = secretions; rub = pleura; grunting = distress

Topic	Key points
Report now	SpO_2 <92%, stridor, asymmetrical expansion, worsening dyspnea, altered LOC/restlessness

Self-Check ADDED

1. A patient smoked 1½ packs a day for 20 years. Pack-years? *Answer: 30.*
2. Where should the nurse listen to assess the right middle lobe? *Answer: Anteriorly/laterally on the right chest.*
3. A child has a high-pitched sound on inspiration and is drooling. Priority? *Answer: Stridor = upper airway obstruction; call for help immediately.*
4. Popping sounds on inspiration in the lung bases suggest what? *Answer: Crackles, from fluid in the alveoli.*
5. A patient with COPD has an AP:lateral ratio of 1:1. What is this called? *Answer: Barrel chest.*
6. New restlessness and anxiety in a patient with pneumonia should make the nurse think of what? *Answer: Hypoxia; check SpO_2 and report.*

Notes on Sources, Additions & Clarifications

Corrections:

1. **Substernal retractions.** The slide text says "Indentation *above* sternum," which is actually the suprasternal location. The slide's own diagram labels substernal retractions as "pulling in of the skin **below** the sternum," which is correct. I used the diagram.

From the Exam 1 study guide (not spelled out on the slides):

- Module 3 question count (20–22) and the topic list.
- The 1:2 (normal) vs. 1:1 (barrel chest) AP ratio. The slides say only "increased AP diameter."
- Pleuritic chest discomfort as a symptom question.
- Acute stridor and asymmetrical chest expansion as red flags to escalate (the slides list SpO_2 <92%, pain, worsening dyspnea, and LOC changes).

Added information (not in the slides):

- Names of the accessory muscles.
- Pack-year practice math.
- How to palpate chest expansion (thumbs at the 10th rib).
- Inspiration/expiration timing for normal breath sounds (it is in the slide's image but not its text) and the bronchovesicular landmarks.
- Rhonchi may clear with coughing; grunting is especially seen in infants.
- The self-check questions.

Clarified:

- Why pediatric and older adult differences matter clinically.
- A normal breath sound in the wrong location is abnormal.

Other: The image-only lung anatomy slides were replaced by the lobes diagram. The lab list comes from the slide notes.